

A General U-Statistic Framework for High-Dimensional Multiple Change-Point Analysis

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SUMMARY: High-dimensional change-point analysis is essential in modern statistical inference. However, existing methods are often designed either for specific parameters (e.g., mean or variance) or for particular tasks (e.g., testing or estimation), making them difficult to generalize. Moreover, they typically rely on restrictive distributional assumptions, limiting their robustness to heavy-tailed data. We propose a unified framework for testing, estimating, and inferring multiple change points in high-dimensional data. Our approach leverages a two-sample U-statistic within a moving window, allowing flexible kernel function selection to accommodate structural changes in general parameters such as variance changes or robust statistics. For testing, we develop an ℓ_∞ -norm-based statistic with a high-dimensional multiplier bootstrap procedure, achieving minimax-optimal power under sparse alternatives. For estimation, we construct an initial estimator for the change-point number and locations and refine it using the U-statistic Projection Refinement Algorithm (U-PRA), attaining minimax-optimal localization rates. We further derive the asymptotic distribution of refined estimators, enabling valid confidence interval construction. Extensive numerical experiments demonstrate the superior performance of our method across various settings, including heavy-tailed distributions. Applications to genomic copy number variation data highlight its practical utility.

KEY WORDS: Bootstrap; Efficient computation; Heavy tail; Structure change; U-statistics

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1. Introduction

Modern statistical applications frequently involve high-dimensional data sequences, where the dimensionality substantially exceeds the sample size. Due to the complex nature of data generation processes, heterogeneity is a prevalent characteristic in such datasets. These structural changes, occurring at unknown locations within the observed data sequence, are referred to as change points. Detecting and identifying change points is crucial in many real-world problems. In biological and biomedical studies, for instance, change-point analysis has been widely used to uncover structural shifts in gene expression profiles (Zhang et al. (2010)), detect abrupt transitions in neural activity patterns (Safikhani and Shojaie (2022)), and identify critical stages in disease progression (Ma et al. (2024)). Such insights are essential for understanding complex biological processes and improving early diagnosis strategies.

To formalize the problem, let $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a d -dimensional random vector, and let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be an independent sequence of observations drawn from $\mathbf{X}$, where $\mathbf{X}_t = (X_{t,1}, \dots, X_{t,d})^\top$; $t = 1, \dots, n$. Suppose $\boldsymbol{\theta} \in \mathbb{R}^d$ represents an underlying parameter of interest, and consider the following multiple change-point model:

$$\boldsymbol{\theta}_1 = \dots = \boldsymbol{\theta}_{\gamma_1} \neq \boldsymbol{\theta}_{\gamma_1+1} = \dots = \boldsymbol{\theta}_{\gamma_2} = \dots = \boldsymbol{\theta}_{\gamma_{M_0}} \neq \boldsymbol{\theta}_{\gamma_{M_0}+1} = \dots = \boldsymbol{\theta}_n, \quad (1)$$

where $0 < \gamma_1 < \dots < \gamma_{M_0} < n$ are the unknown change-point locations such that $\boldsymbol{\theta}_t = \boldsymbol{\theta}^{(m)}$ for $\gamma_{m-1} < t \leq \gamma_m$ with $1 \leq m \leq M_0 + 1$. For notational convenience, we define $\gamma_0 = 0$ and $\gamma_{M_0+1} = n$. Given this model, high-dimensional change-point analysis typically involves three primary research questions: (1) detecting the presence of change points, (2) estimating multiple change points, and (3) conducting statistical inference for the estimated change-point locations.

The first problem, single change-point detection assuming $M_0 \leq 1$, has been extensively studied, particularly in the context of detecting mean shifts with $\boldsymbol{\theta}_t = \mathbb{E}[\mathbf{X}_t]$. In high-dimensional settings, the change may occur in either a sparse subset of components or across

a dense set of coordinates with small individual signals. Consequently, recent research has focused on strategies to efficiently aggregate information from affected components to enhance detection power. Existing detection methods are typically categorized into two classes: those that assume prior knowledge about the sparse or dense nature of the alternative hypothesis, and those that employ adaptive procedures to determine the alternative structure in a data-driven manner. For instance, sparse change-point detection using ℓ_∞ -norm aggregation was explored in Jirak (2015); Yu and Chen (2021), while dense alternatives leveraging ℓ_2 -norm aggregation were studied in Wang et al. (2022). A more general adaptive framework combining both perspectives was developed in Cho (2016); Liu et al. (2020); Zhang et al. (2022); Wang and Feng (2023). For a more comprehensive discussion, see the review by Liu et al. (2022). However, most of these methods assume that the error terms in the data are sub-Gaussian, sub-exponential or require stringent moment conditions. When these assumptions are violated, such as in the presence of heavy-tailed distributions or outliers, mean-based statistics can suffer from inflated false detection rates and reduced power, limiting their practical reliability.

Beyond detecting a single change point, multiple change-point estimation with $M_0 \geq 1$ aims to determine both the number and locations of structural breaks. Two dominant approaches for this problem include segmentation-based and penalization-based methods. The first category extends single change-point detection techniques by incorporating recursive partitioning algorithms, such as binary segmentation (Liu et al. (2020)), wild binary segmentation (WBS) (Zhang et al. (2022)), and seed binary segmentation. While computationally efficient, these methods often lack theoretical guarantees or impose specific requirements on the strength of the change-point signal and suffer from multiple testing issues, making it difficult to control false discovery rates. The second category treats multiple change points as unknown parameters and estimates them via penalized optimization techniques, such

as fused LASSO. Recently, Safikhani and Shojaie (2022) considered joint structural break detection and parameter estimation in high-dimensional nonstationary VAR models. While these methods offer structured estimators, they generally assume that change points exist a priori and do not provide a formal hypothesis testing framework for their presence. Moreover, most of these methods are designed primarily for mean shifts and are not readily extendable to other parameters such as variance changes or robust statistics.

The third major challenge is to conduct statistical inference for estimated change points. While numerous methods have been developed for detecting and estimating change points, relatively few offer confidence intervals or theoretical guarantees for their estimates. The difficulty in constructing confidence intervals lies in establishing optimality and characterizing the asymptotic distribution of change-point estimators. Existing results are largely limited to univariate settings, including recent studies by Eichinger and Kirch (2018) as well as classical work by Bai and Perron (1998). Extending these methods to high-dimensional settings remains particularly challenging, and only recent progress has been made in this direction. Moreover, the construction of confidence intervals for change points has primarily focused on mean vector parameters, with most approaches relying on either an ℓ_2 -loss-based framework (Kaul et al. (2021)) or mean-based CUSUM statistics (Chen et al. (2022)). These methodologies are inherently difficult to generalize to broader parameter settings and typically assume that the data follow a light-tailed distribution. Consequently, when applied to heavy-tailed data, these methods often yield unreliable confidence intervals.

Despite significant progress in high-dimensional change-point analysis, several challenges remain. First, most existing methods are designed for mean-based change detection and rely on strong distributional assumptions, limiting their applicability when the underlying data deviate from Gaussianity. Second, while research on multiple change-point estimation has advanced, many existing techniques either lack theoretical guarantees or do not provide a

formal hypothesis testing framework for determining the presence of change points. Moreover, these methods are often tailored to detecting mean shifts and struggle to generalize to other structural changes, such as variance shifts or robust statistics, particularly in the presence of heavy-tailed distributions or outliers. Beyond estimation, statistical inference for multiple change points remains an open challenge. Confidence interval construction has predominantly focused on mean vector parameters and often imposes restrictive assumptions on the error distribution. Many existing methods tend to break down under heavy-tailed data or in the presence of outliers, making them unreliable in broader applications. Furthermore, extending current inference frameworks to accommodate more general parameters remains a significant hurdle. This limitation highlights the need for a more general and statistically rigorous framework that extends beyond mean-based change point detection to capture broader forms of structural heterogeneity. To overcome these challenges, we propose a novel methodology for detecting, estimating, and inferring multiple change points in high-dimensional data. Our contributions can be summarized as follows.

1. **Multiple Change-Point Testing:** We develop a multiple change-point testing framework based on a two-sample U-statistic within a moving window setting, incorporating an ℓ_∞ -norm-based test statistic tailored for sparse alternatives. By appropriately selecting the kernel function, our method generalizes beyond mean shifts to detect changes in variance, robust mean, and robust variance. To approximate the limiting distribution of the testing statistic, we construct a high-dimensional multiplier bootstrap method that effectively captures the covariance structure among data components and mitigates dependencies induced by the moving window mechanism. Theoretically, we establish that our proposed test controls the significance level under the null hypothesis and achieves asymptotic power one under the alternative, even allowing the dimensionality d to grow exponentially with the sample size n . Notably, under sparse alternatives, our method attains minimax-optimal detection rates if the

bandwidth parameter is chosen properly. Our test imposes minimal moment conditions on the data by shifting regularity conditions onto the kernel function, enabling robust change-point detection in heavy-tailed distributions. By applying a robust kernel, such as the Wilcoxon-Mann-Whitney kernel, our method extends change-point testing to a robustness framework, a direction that, to our knowledge, remains unexplored in the existing literature.

2. Optimal Estimation of Change-Point Locations and Numbers: Inspired by Eichinger and Kirch (2018) in low-dimensional settings, we construct an initial estimator for the number and locations of change points in high-dimensional scenarios. We theoretically demonstrate that under mild model assumptions, this estimator correctly identifies the number with probability tending to one, with location estimates achieving near-optimal convergence rates, up to a logarithmic factor $\log(d)$. To further refine the estimation, we introduce a U-statistic projection refinement algorithm (U-PRA). This method first identifies the subset of components at each change point with high probability and then projects the test statistic onto this subspace along the U-statistic's direction, reducing the problem to a univariate change-point estimation. A local search then fine-tunes the final estimates. We rigorously establish that the refined estimators attain the minimax-optimal convergence rate, which provides optimal change-point estimation not only for mean vectors but also for variance and robust statistics, thereby solving a broad class of change-point estimation problems. Our simulations demonstrate that even in heavy-tailed settings with outliers, our method remains accurate and robust, significantly outperforming existing techniques.

3. Confidence Interval Construction for Multiple Change Points: Building on the refined estimation in the second step, we derive the asymptotic distribution of each change-point estimator within the two-sample U-statistic framework, establishing that it follows the $\arg \max$ of a drifted weighted Brownian motion process. This result enables the construction of valid confidence intervals for individual change points as their signal

strength approaches zero. Additionally, we prove that the limiting distributions of different change-point estimators are asymptotically independent, allowing us to construct joint confidence intervals for multiple change points. Importantly, since our framework is based on U-statistics, the limiting distribution exhibits a novel form involving weighted sums of three independent Brownian motions, a result established here for the first time in high-dimensional change-point analysis. By appropriately selecting the kernel function, our method extends confidence interval construction beyond mean changes to variance, robust mean, and robust variance. Our numerical experiments confirm that even in heavy-tailed settings, our constructed confidence intervals achieve coverage probabilities close to their nominal levels, making a significant advancement in high-dimensional change-point inference. Finally, we conduct extensive numerical studies validating both our theoretical results and methodological effectiveness. Our analysis on genomic copy number variation and S&P 500 datasets further highlights the practical utility of our proposed framework.

The rest of this paper is organized as follows. In Section 2, we introduce our methodology for multiple change point testing and introduce a multiplier bootstrap based procedure for approximating the limiting null distribution in Section 3. The estimation and inference of multiple change points are given in Section 4. In Sections 5 and 6, numerical studies and real data applications are investigated. Discussions are given in Section 7. The supplementary materials provide proofs of the theoretical results and additional numerical results.

We end this section with some notations. Let $\mathcal{X} = \{\mathbf{X}_1, \dots, \mathbf{X}_n\}$. We define the ℓ_p norm as $\|\mathbf{v}\|_p = (\sum_{j=1}^d |v_j|^p)^{1/p}$ for $\mathbf{v} = (v_1, v_2, \dots, v_d)^\top \in \mathbb{R}^d$. For $p = \infty$, $\|\mathbf{v}\|_\infty = \max_{1 \leq j \leq d} |v_j|$. For $p = 0$, $\|\mathbf{v}\|_0 := \#\{j : v_j \neq 0\}$, where $\#\{\mathcal{S}\}$ denotes the cardinality of a set $\mathcal{S}$. For two real numbered sequences a_n and b_n , we set $a_n = O(b_n)$ if there exists a constant C such that $|a_n| \leq C|b_n|$ for a sufficiently large n ; $a_n = o(b_n)$ if $a_n/b_n \rightarrow 0$ as $n \rightarrow \infty$; $a_n \asymp b_n$ if there exists constants c and C such that $c|b_n| \leq |a_n| \leq C|b_n|$ for a sufficiently large n . For

a sequence of random variables (r.v.s) $\{\xi_1, \xi_2, \dots\}$, we set $\xi_n \rightarrow \xi$ if ξ_n converges to ξ in probability as $n \rightarrow \infty$. We also denote $\xi_n = o_p(1)$ if $\xi_n \rightarrow 0$.

2. Testing existence of multiple change points

In this section, we present our methodology for the multiple change point detection. Before presenting our methodology, we introduce some model settings. Let $h(x, y) \in \mathbb{R}^1$ be some user specified two-sample kernel which is antisymmetric in the sense that $h(y, x) = -h(x, y)$. In the following sections, for $\mathbf{x} = (x_1, \dots, x_d)$ and $\mathbf{y} = (y_1, \dots, y_d)$, we set $\mathbf{h}(\mathbf{x}, \mathbf{y}) := (h(x_1, y_1), \dots, h(x_d, y_d)) \in \mathbb{R}^d$. Let $\theta_j^{(m)} := \mathbb{E}h(X_{\gamma_m, j}, X_{\gamma_{m+1}, j})$ be the signal jump for $m = 1, \dots, M_0$ and $j = 1, \dots, d$ and define $\boldsymbol{\theta}^{(m)} := \mathbb{E}\mathbf{h}(\mathbf{X}_{\gamma_m}, \mathbf{X}_{\gamma_{m+1}})$ as the $\mathbb{R}^d$ -dimensional signal jump at the γ_m -th change point. It is worth noting that in many ultrahigh-dimensional change-point detection problems, the associated kernel functions are often antisymmetric. Moreover, by appropriately selecting some kernel functions, our framework can recover several existing high-dimensional change-point detection methods for specific parameters. For instance, setting $h(x, y) = y - x$ leads to $\theta_j^{(m)} := \mathbb{E}(X_{\gamma_{m+1}, j} - X_{\gamma_m, j})$ representing the mean shift between segments. In subsequent sections, we will provide further examples to illustrate the versatility and broad applicability of our approach. We set $\Pi_m = \{j \in \{1, \dots, d\} : \theta_j^{(m)} \neq 0\}$ as the set of coordinates with a change point at the change point γ_m and $s_m = |\Pi_m|$ as the total number of coordinates.

Our first goal is to present a new test to determine the presence of multiple breaks. To formally establish the framework, define the signal jump at time t as:

$$\boldsymbol{\theta}_t := \mathbb{E}\mathbf{h}(\mathbf{X}_t, \mathbf{X}_{t+1}), \quad \text{for } t = 1, \dots, n-1, \quad (2)$$

where $\mathbf{h}(\mathbf{x}, \mathbf{y})$ is a user-specified kernel function that quantifies the structural difference between two consecutive observations. Based on this definition, we formulate the hypothesis

testing problem as follows:

$$\begin{aligned} \mathbf{H}_0 : \boldsymbol{\theta}_t = 0, \quad \forall t = 1, \dots, n-1, \\ \mathbf{H}_1 : \exists \gamma_1 < \gamma_2 < \dots < \gamma_{M_0}, \text{ such that } \boldsymbol{\theta}_{\gamma_m} \neq 0, \quad \forall m = 1, \dots, M_0. \end{aligned} \quad (3)$$

Under the null hypothesis $\mathbf{H}_0$, the entire sequence remains stable, meaning that no significant structural changes occur. In contrast, under the alternative hypothesis $\mathbf{H}_1$, there exist at least one or more change-points γ_m where the underlying process exhibits abrupt shifts.

Note that our testing method allows the number of change points M_0 scale with the sample observations n which covers the single change point testing as a special case when $M_0 = 1$. Our method is based on the moving window based procedure proposed in Eichinger and Kirch (2018). Specifically, let $k \in \{1, \dots, n\}$ be a candidate search location and $1 \leq G \leq n$ be some bandwidth. For each search window $[s_k, e_k] := \{k - G + 1, k - G + 2, \dots, k, k + 1, \dots, k + G\}$ and for each coordinate $j \in \{1, \dots, d\}$, we define the following two-sample U-statistic based testing statistics:

$$T_j(k) := \frac{1}{G^{3/2}} \sum_{t_1=k-G+1}^k \sum_{t_2=k+1}^{k+G} h(X_{t_1,j}, X_{t_2,j}), \quad \text{for } j = 1, \dots, d. \quad (4)$$

Intuitively, the bandwidth G should be sufficiently large to ensure that each search interval contains an adequate number of observations. However, G should not be so large as to allow more than one change point within any given scanning interval. Details on this condition will be provided in the theoretical analysis section. See Assumption (B.1).

Next, we first explain why the aforementioned test statistic can effectively detect multiple change points in the data. It can be proven that by selecting an appropriate bandwidth G , the sliding window $[s_k, e_k]$ contains at most one change point, and these sliding windows with a single change point determine the testing and estimation of the change point. Suppose the bandwidth G is chosen properly such that there exists only one change point, say γ_m , in $[s_k, e_k]$. To further elucidate this issue, we first introduce Hoeffding's decomposition for the kernel $h(x, y)$. In particular, for each coordinate $j = 1, \dots, d$ and change point γ_m , by

Hoeffding's decomposition, we can decompose $h(x, y)$ as follows:

$$h(X_{t_1,j}, X_{t_2,j}) = \theta_j^{(m)} + h_{1,j}(X_{t_1,j}) + h_{2,j}(X_{t_2,j}) + g_j(X_{t_1,j}, X_{t_2,j}), \quad (5)$$

where $\theta_j^{(m)} = \mathbb{E}h(X_{\gamma_m,j}, X_{\gamma_m+1,j})$, and for any $\gamma_{m-1} < t_1 \leq \gamma_m, \gamma_m + 1 \leq t_2 \leq \gamma_{m+1}$, the three terms $h_{1,j}(X_{t_1,j})$, $h_{2,j}(X_{t_2,j})$ and $g_j(X_{t_1,j}, X_{t_2,j})$ are defined as

$$\begin{aligned} h_{1,j}(X_{t_1,j}) &= \mathbb{E}[h(X_{t_1,j}, X_{t_2,j})|X_{t_1,j}] - \theta_j^{(m)}, \quad h_{2,j}(X_{t_2,j}) = \mathbb{E}[h(X_{t_1,j}, X_{t_2,j})|X_{t_2,j}] - \theta_j^{(m)}, \\ \text{and } g_j(X_{t_1,j}, X_{t_2,j}) &= h(X_{t_1,j}, X_{t_2,j}) - \theta_j^{(m)} - h_{1,j}(X_{t_1,j}) - h_{2,j}(X_{t_2,j}). \end{aligned} \quad (6)$$

Hence, for any $k \in [s_k, e_k]$ with only one change point $\gamma_m \in [s_k, e_k]$, $T_j(k)$ can have the following explicit form:

$$T_j(k) = \begin{cases} \delta_j^{(m)}(k) + \underline{R}_{1,j}(k) + \underline{R}_{2,j}(k) + R_{3,j}(k), & \gamma_{m-1} < k \leq \gamma_m, \\ \delta_j^{(m)}(k) + \overline{R}_{1,j}(k) + \overline{R}_{2,j}(k) + R_{3,j}(k), & \gamma_m < k \leq \gamma_{m+1}, \end{cases} \quad (7)$$

where

$$\delta_j^{(m)}(k) = \frac{G(k + G - \gamma_m)}{G^{3/2}} \theta_j^{(m)} \mathbf{1}\{\gamma_{m-1} < k \leq \gamma_m\} + \frac{G(\gamma_m - k + G)}{G^{3/2}} \theta_j^{(m)} \mathbf{1}\{\gamma_m < k \leq \gamma_{m+1}\}$$

is the called signal jump function and

$$\begin{aligned} \underline{R}_{1,j}(k) &:= \frac{k + G - \gamma_m}{G^{3/2}} \sum_{t_1=k-G+1}^k h_{1,j}(X_{t_1,j}) + \frac{G}{G^{3/2}} \sum_{t_2=\gamma_m+1}^{k+G} h_{2,j}(X_{t_2,j}), \\ \underline{R}_{2,j}(k) &:= \frac{1}{G^{3/2}} \sum_{t_1=k-G+1}^k \sum_{t_2=k+1}^{\gamma_m} h(X_{t_1,j}, X_{t_2,j}), \quad \text{for } \gamma_{m-1} < k \leq \gamma_m; \\ \overline{R}_{1,j}(k) &:= \frac{G}{G^{3/2}} \sum_{t_1=k-G+1}^{\gamma_m} h_{1,j}(X_{t_1,j}) + \frac{\gamma_m - k + G}{G^{3/2}} \sum_{t_2=k+1}^{k+G} h_{2,j}(X_{t_2,j}), \\ \overline{R}_{2,j}(k) &:= \frac{1}{G^{3/2}} \sum_{t_1=\gamma_m+1}^k \sum_{t_2=k+1}^{k+G} h(X_{t_1,j}, X_{t_2,j}), \quad \text{for } \gamma_m < k \leq \gamma_{m+1}; \\ R_{3,j}(k) &:= \frac{1}{G^{3/2}} \sum_{t_1=k-G+1}^k \sum_{t_2=k+1}^{k+G} g_j(X_{t_1,j}, X_{t_2,j}), \quad \text{for } \gamma_{m-1} < k \leq \gamma_{m+1}. \end{aligned}$$

Note that $\underline{R}_{1,j}(k)$ and $\overline{R}_{1,j}(k)$ are the leading term (random noise) of the Hoeffding's decomposition, $R_{3,j}(k)$ is the negligible term (random bias), and $\underline{R}_{2,j}(k)$ and $\overline{R}_{2,j}(k)$ involve data with no change point. Based on the above decomposition, we see that $|\delta_j^{(m)}(k)|$ obtains its maximum value at the true change point γ_m . Moreover, under some regular conditions, we can prove that

$$\max_{1 \leq j \leq d} \max_{G \leq k \leq n-G} \left(|\underline{R}_{1,j}(k)| + |\underline{R}_{2,j}(k)| + |\overline{R}_{1,j}(k)| + |\overline{R}_{2,j}(k)| + |R_{3,j}(k)| \right) = O_p(\sqrt{\log(nd)}).$$

The above observation suggests that by imposing certain signal-to-noise ratio requirements on the change-point signals $\theta_j^{(m)}$, we can ensure that the signal function $\delta_j^{(m)}(k)$ dominates the random noise and biases, thereby enabling accurate detection and identification of change points. To further illustrate the novelty of our proposed methodology, in Web Appendix A, we provide several specific examples of kernel functions $h(x, y)$ used in the construction of the two-sample U-statistics.

Let $\mathbf{T}(k) = (T_1(k), \dots, T_d(k))^\top$. Note that we aim to propose a powerful testing procedure that is tailored to sparse alternatives. Hence, we consider the ℓ_∞ -norm aggregation of $\mathbf{T}(k)$ which has been used in Jirak (2015); Yu and Chen (2021, 2022). Specifically, our final testing statistic is defined as follows:

$$W = \max_{G \leq k \leq n-G} \max_{1 \leq j \leq d} |T_j(k)| = \max_{G \leq k \leq n-G} \|\mathbf{T}(k)\|_\infty. \quad (8)$$

Since W is constructed by adopting the ℓ_∞ norm, it is more sensitive to sparse signals with strong perturbations on a small number of coordinates. More importantly, we do not employ the CUSUM statistic; instead, we utilize information of two-sample U statistics that constructs the testing statistic. This ensures that the test may not use the data points directly and may maintain high power while also being robust to heavy-tailed data and outliers.

Given the significance level α , we reject the null hypothesis once W exceeds a critical value, say $c_{\alpha, W}$. In the following section, we propose a technique to obtain the critical values $c_{\alpha, W}$ which is based on the multiplier bootstrap such that the testing can control the type I error while being consistent under sparse alternatives.

3. Testing using bootstrap quantiles

In the previous section, we introduce the ℓ_∞ -norm based testing statistic W in (8). It is difficult to obtain its limiting distribution directly as n and d go to infinity. Under some extremely stringent model assumptions, we may derive that the theoretical asymptotic distribution of this statistic is the Gumbel distribution. However, this typically imposes

very strong requirements on the covariance structure of the data, which are often difficult to satisfy in practical applications. More importantly, the method relying on approximating the statistic's distribution with a Gumbel distribution is known to have slow convergence rates using such asymptotic results. To overcome this problem, in this section, we introduce the multiplier bootstrap method for approximating the limiting distribution of W . Bootstrap methods for U -statistics have been developed in low dimensions (Bücher and Kojadinovic, 2016) and extended to high dimensions (Chernozhukov et al., 2013, 2017; Jirak, 2015; Liu et al., 2020). Unlike existing literature, we propose a novel multiplier bootstrap procedure using two-sample U -statistics with moving windows for multiple change point detection. Let $e_1^b, \dots, e_n^b$ be i.i.d $N(0, 1)$ random variables with $b = 1, \dots, B$. Then the b -th bootstrap version of the coordinate based testing statistic $T_j(k)$ can be defined as:

$$T_j^b(k) = \frac{1}{G^{3/2}} \sum_{t_1=k-G+1}^k \sum_{t_2=k+1}^{k+G} (e_{t_1} + e_{t_2}) h(X_{t_1,j}, X_{t_2,j}), \quad \text{for } j = 1, \dots, d. \quad (9)$$

Accordingly, the b -th bootstrap version of the final testing statistic is defined as:

$$W^b = \max_{G \leq k \leq n-G} \max_{1 \leq j \leq d} |T_j^b(k)| = \max_{G \leq k \leq n-G} \|\mathbf{T}^b(k)\|_\infty.$$

Given the significance level α , let $c_{\alpha,W} := \inf\{t \in \mathbb{R} : \mathbb{P}(W \leq t) \geq 1 - \alpha\}$ be the theoretical critical value for W . Based on the bootstrap samples $\{W^1, \dots, W^B\}$, we estimate $c_{\alpha,W}$ by

$$\hat{c}_{\alpha,W} = \inf \left\{ t : \frac{\sum_{b=1}^B \mathbb{I}\{W^b \leq t\}}{B} \geq 1 - \alpha \right\}. \quad (10)$$

Hence, based on the estimated critical values $\hat{c}_{\alpha,W}$, we define our test as follows

$$\Psi_{\alpha,W} = \mathbb{I}\{W \geq \hat{c}_{\alpha,W}\}. \quad (11)$$

For W in (8), we reject $\mathbf{H}_0$ if and only if $\Psi_{\alpha,W} = 1$.

The multiplier bootstrap procedure captures the underlying covariance structure of the data in an automatic way. More importantly, as a result of the moving window procedure, $\{\mathbf{T}(k), k = G, \dots, n - G\}$ possesses a complex correlation structure across different scanning positions. We prove that the bootstrap-based test statistic can approximate this correlation structure in a fully data-driven manner, thereby circumventing the necessity

of first estimating the covariance matrix and then generating the corresponding Gaussian random variables for approximation as used in Chen et al. (2022). Moreover, the bootstrap method is straightforward to implement, requiring no hyperparameter tuning or additional estimation of high-dimensional parameters. Furthermore, when estimating multiple change points, we can utilize the bootstrap-based critical values as thresholds for identifying change-point intervals, thereby avoiding the need to manually specify a threshold as used in Wang and Samworth (2018). Additionally, the proposed bootstrap procedure is not limited to mean parameters; it is also well-suited for detecting variance change points and robust mean shifts, highlighting its high flexibility and broad applicability.

Next, we provide theoretical results for the bootstrap based procedure under $\mathbf{H}_0$. Let $\mathbf{X} = (X_1, \dots, X_d)^\top \sim F(\mathbf{x})$ under $\mathbf{H}_0$ and $\mathbf{X}' = (X'_1, \dots, X'_d)^\top$ be the independent copy of $\mathbf{X}$. Under $\mathbf{H}_0$, for the kernel $h(x, y)$, consider the following Hoeffding's decomposition

$$h(X_j, X'_j) = 0 + h_{1,j}(X_j) + h_{2,j}(X'_j) + g_j(X_j, X'_j), \quad j = 1, \dots, d, \quad (12)$$

where $h_{1,j}(X_j) := \mathbb{E}[h(X_j, X'_j)|X_j]$, $h_{2,j}(X'_j) := \mathbb{E}[h(X_j, X'_j)|X'_j]$, and $g_j(X_j, X'_j) = h(X_j, X'_j) - h_{1,j}(X_j) - h_{2,j}(X'_j)$. Note that due to the antisymmetry property of $h(x, y)$, we have $h_{2,j}(x) = -h_{1,j}(x)$. The following Assumptions (A.1) – (A.3) impose some moment conditions on the kernel. Specifically,

Assumption (A.1): There is some constant b such that $\mathbb{E}(h_{1,j}(X_j))^2 \geq b$ for all $j = 1, \dots, d$.

Assumption (A.2): There exists some constant D such that $\|h_{1,j}(X_j)\|_{\psi_1} \leq D$ for all $j = 1, \dots, d$, where for any random variable X , $\|X\|_{\psi_1}$ is defined as $\|X\|_{\psi_1} := \inf \{C > 0 : \mathbb{E}\psi_1(|X|/C)\} \leq 1$ with $\psi_1(x) := \exp(x) - 1$.

Assumption (A.3): There exists some constant D such that $\mathbb{E}|h_{1,j}(X_j)|^{2+\ell} \leq D^\ell$ with $\ell = 1, 2$ for all $j = 1, \dots, d$.

Assumption (A.4) For the bandwidth parameter G , we require

$$\frac{\log^7((n - 2G + 1)d)}{G} \rightarrow 0, \text{ as } (n, d) \rightarrow \infty.$$

Assumptions (A.1) – (A.4) are crucial for proving the results of Gaussian approximations for the ℓ_∞ -norm-based testing statistic. Assumption (A.1) requires that the two sample kernel based U -statistics are non-degenerate. Assumption (A.2) requires that $h_{1,j}(X_j)$ follows sub-exponential distributions. Many bounded kernels such as Wilcoxon can satisfy this condition. Assumption (A.3) requires that $h_{1,j}(X_j)$ has bounded third and forth moments. Lastly, Assumption (A.4) characterizes the scaling relationship between (G, n, d) which allows the data dimension d can grow exponentially with n as long as $G \gg \log^7((n - 2G + 1)d)$ holds. Bases on the above assumptions, the following Theorem 1 justifies the validity of our proposed bootstrap procedure.

THEOREM 1: *Suppose Assumptions (A.1) – (A.4) hold. Under $\mathbf{H}_0$, we have*

$$\sup_{z \in (0, \infty)} |\mathbb{P}(W \leq z) - \mathbb{P}(W^b \leq z | \mathcal{X})| = o_p(1), \text{ as } n, d \rightarrow \infty. \quad (13)$$

Theorem 1 shows that we can uniformly approximate the distribution of W using that of W^b . As a corollary, the following Corollary 1 shows that our proposed new test can control the Type I error asymptotically for any given pre-specified significant level α .

COROLLARY 1: *Assume Assumptions (A.1)–(A.4) hold. Under $\mathbf{H}_0$, we have $\mathbb{P}(\Psi_{\alpha, W} = 1) \rightarrow \alpha$, as $n, d, B \rightarrow \infty$.*

The proof of Theorem 1 relies on the essential modifications of high dimensional Gaussian approximation theory of Chernozhukov et al. (2017) to the two-sample U-statistic-based multiple change point detection with moving window. The proof mainly proceeds in three steps. More detailed proof is presented in Appendix B.1 of supplementary materials.

After analyzing the size, we discuss the power properties. To this end, we need some additional notations. Recall $\Pi_m := \{j \in \{1, \dots, d\} : \theta_j^{(m)} \neq 0\}$ as the set of coordinates having a change point at γ_m . For each $m = 1, \dots, M_0$, define the signal jump vector $\boldsymbol{\theta}^{(m)} := (\theta_1^{(m)}, \dots, \theta_d^{(m)})^\top$. Define $\boldsymbol{\theta}^\diamond = \max_{1 \leq m \leq M_0} \|\boldsymbol{\theta}^{(m)}\|_\infty$ be the maximum signal jump among all

change points and let $\Delta := \min_{1 \leq m \leq M_0} (\gamma_{m+1} - \gamma_m)$ be the minimum segmentation length between change points with $\gamma_{M_0+1} = n$. Next, we present some conditions for $\mathbf{H}_1$.

Assumption (B.1): For the bandwidth parameter G , we require

$$\Delta \geq 2G, \text{ and } \frac{\log^2(nd) \log^2(Gd)}{G} \rightarrow 0 \text{ as } (n, d) \rightarrow \infty.$$

Additionally, suppose there exist some constants such that $\max_{1 \leq m \leq M_0} \|\boldsymbol{\theta}^{(m)}\|_\infty \leq C^*$.

Assumption (B.2): There exists some constants D^ℓ such that $\mathbb{E}|h(X_{\gamma_m, j}, X_{\gamma_{m+1}, j}) - \theta_j^{(m)}|^{2+\ell} \leq D^\ell$ with $\ell = 1, 2$ for all $j = 1, \dots, d$ and $m = 1, \dots, M_0$.

Assumption (B.3): There exists some constant D such that $\|h(X_{\gamma_m, j}, X_{\gamma_{m+1}, j}) - \theta_j^{(m)}\|_{\psi_1} \leq D$ for all $j = 1, \dots, d$ and $m = 1, \dots, M_0$.

Assumption (B.1) requires that the bandwidth should not be too large in the sense that $G \leq \Delta/2$ such that there is at most one change point in any candidate moving window $[k-G+1, k+G]$. Moreover, the bandwidth is required to have an order $G \gg \log^2(nd) \log^2(Gd)$ so that the large sample property holds. See also Eichinger and Kirch (2018). Assumptions (B.2) and (B.3) make some similar moment conditions on the kernel to Assumptions (A.2) and (A.3). Notably, our moment condition assumptions are primarily imposed on the kernel rather than the data itself. Therefore, for many heavy-tailed datasets, the theoretical framework of the model remains applicable as long as an appropriate kernel function is chosen.

With the above notations and some regularity conditions, the following Theorem 2 shows that we can reject the null hypothesis of no change point with overwhelming probability.

THEOREM 2: *Suppose Assumptions (B.1) – (B.3) hold. If $\boldsymbol{\theta}^\diamond$ satisfies*

$$\sqrt{G} \times \boldsymbol{\theta}^\diamond \geq C_0 \left(\sqrt{2 \log(dn)} + \sqrt{2 \log(\alpha^{-1})} \right), \quad (14)$$

under $\mathbf{H}_1$, we have $\mathbb{P}(\Phi_{\alpha, W} = 1) \rightarrow 1$, as $n, d, B \rightarrow \infty$, where C_0 is a large enough universal positive constant not depending on n or d .

Theorem 2 demonstrates that with probability tending to one, our proposed new test can detect the existence of multiple change points as long as the corresponding signal to noise

ratio satisfies (14). Specifically, considering Assumption (B.1) with $\Delta \geq 2G$, if the bandwidth parameter is chosen properly with $\Delta \approx G$ it mainly requires that

$$\max_{1 \leq m \leq M_0} \|\boldsymbol{\theta}^{(m)}\|_\infty \geq \sqrt{\log(nd)/\Delta}. \quad (15)$$

Hence, with a larger signal jump and larger segmentation length, it is more likely to trigger a rejection of the null hypothesis. From the above conditions, we can also observe that to detect multiple change points, it is sufficient for the signal-to-noise ratio condition of just one of the change points to meet the requirements. This is fundamentally different from the conditions for change point estimation discussed later.

4. Estimation and inference of break points

After introducing our test statistic for Problem 3, we next present our methodology for identifying the unknown change point number, locations, as well as constructing their valid confidence intervals. Our estimation algorithms proceed into two steps. In the first step, we modify the ideas in Eichinger and Kirch (2018) to the high dimensional settings based on the two sample kernel $h(x, y)$ with ℓ_∞ -norm aggregations.

Step 1: Obtain initial estimator. Recall $\mathbf{T}(k) = (T_1(k), \dots, T_d(k))^\top$ with $T_j(k)$ defined in (4). Consider all pairs of indices $(v_j, w_j) \subset \{G, \dots, n - G\}$ such that

$$\begin{aligned} (1) & \|\mathbf{T}(k)\|_\infty \geq c_{\alpha, W}, \text{ for } k = v_j, \dots, w_j; (2) \|\mathbf{T}(k)\|_\infty < c_{\alpha, W}, \text{ for } k = v_j - 1, w_j + 1, \\ (3) & w_j - v_j \geq \eta G, \text{ for some fixed but arbitrary } 0 < \eta < 1/2, \end{aligned} \quad (16)$$

where $c_{\alpha, W}$ is the α quantile level obtained from the bootstrap procedure in Section 3. Condition 16 is necessary to avoid overestimation by spurious local maxima exceeding the critical value on the boundary between significant and insignificant areas. Based on (16), our initial estimation for change point number and locations are defined as:

$$\hat{M}_0 = \text{number of pairs } (v_j, w_j), \text{ and } \hat{\gamma}_m = \arg \max_{v_m \leq k \leq w_m} \|\mathbf{T}(k)\|_\infty \text{ for } m = 1, \dots, M_0. \quad (17)$$

Note that the principle in (16) was proposed in Eichinger and Kirch (2018) for one-dimensional sequences and recently modified by Chen et al. (2022). Differently, we consider

the high dimensional setting using the ℓ_∞ -norm aggregations for the two-sample U-statistic-based moving window process. In the following, we provide consistency results of the initial estimated change point number and locations. To that end, we need the following conditions on the signal strength $\boldsymbol{\theta}_\diamond = \min_{1 \leq m \leq M_0} \|\boldsymbol{\theta}^{(m)}\|_\infty$ with $\boldsymbol{\theta}^{(m)} := \mathbb{E}\mathbf{h}(\mathbf{X}_{\gamma_m}, \mathbf{X}_{\gamma_m+1})$.

Assumption (C.1) Assume $G \times \boldsymbol{\theta}_\diamond^2 \gg \log(nd)$.

Assumption (C.1) is the so called signal to noise ratio condition for identifying multiple change points. Considering Assumption (B.1) with $\Delta \geq 2G$, it can be equivalently stated as $\Delta \times \boldsymbol{\theta}_\diamond^2 \gg \log(nd)$. Hence, with a larger segmentation length Δ and larger signal strength $\boldsymbol{\theta}_\diamond$, Assumption (C.1) is more likely to be satisfied.

THEOREM 3: (*Consistency of initial estimators*) Suppose Assumptions (B.1)-(B.3) and (C.1) hold. Then we have

- (1) $\mathbb{P}\{\hat{M}_0 = M_0\} \rightarrow 1$, as $n, d \rightarrow \infty$;
- (2) $|\hat{\gamma}_m - \gamma_m| = O_P\left(\frac{\log(nd)}{\|\boldsymbol{\theta}^{(m)}\|_\infty^2}\right)$, for $m = 1, \dots, M_0$.

Theorem 3 (1) shows that the initial procedure can correctly identify the change point number M_0 using the ℓ_∞ -norm aggregation of U-statistic-based moving window process. This is important for the refining procedure in the following step. More importantly, Theorem 3 (2) demonstrates that the initial estimator $\hat{\gamma}_m$ has an estimation error bound of $O_P(\log(nd)/\|\boldsymbol{\theta}^{(m)}\|_\infty^2)$, which is consistent.

Step 2: Obtain refined estimator. Next, we introduce more refined theory of change point estimation. To that end, we propose a U-statistic projection refinement algorithm (U-PRA). This approach initially detects the subset of components at each change point with high probability and subsequently projects the test statistic onto this subspace in the direction of the U-statistic. This transformation simplifies the problem to univariate change-point estimation. Finally, a local search is applied to refine the estimates further. Our objectives are twofold. First, we aim to further optimize the theoretical results of change point estimation in Theorem 3 (2) from $O_P(\log(nd)/\|\boldsymbol{\theta}^{(m)}\|_\infty^2)$ to $O_P(1/\|\boldsymbol{\theta}^{(m)}\|^2)$, which is

known to be theoretical optimality (Verzelen et al. (2023)). Building on this, we construct valid confidence intervals for the positions of multiple change points.

To accomplish the above goal, we need to construct U-statistic-based projection directions. Recall Π_m as the set of coordinates having a change point at γ_m . We construct estimation for Π_m . For each change point γ_m , we estimate its signal jump size for each coordinate by

$$\hat{\theta}_j^m = \frac{1}{G^2} \sum_{t_1=\hat{\gamma}_m-3G/2}^{\hat{\gamma}_m-G/2} \sum_{t_2=\hat{\gamma}_m+G/2}^{\hat{\gamma}_m+3G/2} h(X_{t_1,j}, X_{t_2,j}), \text{ for } j = 1, \dots, d. \quad (18)$$

Note that by Theorem 3, we have proved that with probability tending to one, $|\hat{\gamma}_m - \gamma_m| = o(G)$. Hence, this guarantees that the samples that used to obtain $\hat{\theta}_j^m$ are not mixed. Then, the final estimation for Π_m is defined as

$$\hat{\Pi}_m = \{1 \leq j \leq d : |\hat{\theta}_j^{(m)}| \geq w^+\}. \quad (19)$$

In practice, w^+ is some threshold parameter which can be chosen to be large enough to ensure that we can detect all the jumps with probability tending to one as in Theorem 4.

THEOREM 4: *Suppose Assumptions (B.1)–(B.3) and (C.1) hold. Furthermore, for the threshold parameter w^+ , we require*

$$\begin{aligned} w^+ &\leq \min_{1 \leq m \leq M_0} \min_{1 \leq j \leq d} |\theta_j^{(m)}| - C_2 D \left(\sqrt{\frac{\log(nd)}{G}} + \frac{\log(Gd) \log(nd)}{G} \right) \\ \text{and } w^+ &\geq C_2 D \left(\sqrt{\frac{\log(nd)}{G}} + \frac{\log(Gd) \log(nd)}{G} \right), \end{aligned}$$

for some $C_2 > 0$. Then we have $\mathbb{P}(\hat{\Pi}_m = \Pi_m) \rightarrow 1$.

Using $\hat{\Pi}_m$, instead of aggregating the coordinate with the maximum value, we aggregate coordinates that have a signal jump larger than w^+ . Specifically, for each estimated change point location $\hat{\gamma}_m$ with $|k - \hat{\gamma}_m| \leq G/4$, we define

$$\tilde{T}_j(k) := \frac{1}{G} \sum_{t_1=k-G+1}^k \sum_{t_2=k+1}^{k+G} \hat{\theta}_j^{(m)} h(X_{t_1,j}, X_{t_2,j}), \text{ for } j = 1, \dots, d. \quad (20)$$

Note that $\tilde{T}_j(k)$ can be understood as a projection of the high-dimensional observations $h(X_{t_1,j}, X_{t_2,j})$ according to the direction of $\hat{\theta}^{(m)} = (\hat{\theta}_1^m, \dots, \hat{\theta}_d^m)^\top$. The idea is similar to Wang and Samworth (2018). In addition, in high-dimensional settings, there are other projection-based methods for constructing change point tests, including those by Aston and Kirch

(2018); Chen et al. (2022) and others. However, these methods primarily rely on sample means to determine projection directions. Unlike existing approaches, we have developed U-statistic based projection directions using the general kernel $h(x, y)$ to obtain robust support sets for change points. Based on the aggregated series, the refined change-point locations can be detected as follows:

$$\tilde{\gamma}_m = \arg \max_{|k - \hat{\gamma}_m| \leq G/4} \sum_{j \in \tilde{\Pi}_m} \tilde{T}_j(k). \quad (21)$$

The following theorem shows that after refinemet, our proposed U-PRA can have an optimal convergence rate. Thanks to this result, we are able to construct valid confidence intervals for the positions of change points. To that end, we need more notations. Recall $\Pi_m = \{j \in \{1, \dots, d\} : \theta_j^{(m)} \neq 0\}$ as the coordinates having a change point at γ_m and $s_m = |\Pi_m|$ is the total number. Without loss of generality, we assume $\Pi_m = \{1, \dots, s_m\}$. For each $m = 1, \dots, M_0$ and $j = 1, \dots, d$, define the centralized kernel $\tilde{h}(x, y) = h(x, y) - \theta_j^{(m)}$ and let

$h_{1,j}(X_{\gamma_m,j}) := \mathbb{E}[\tilde{h}(X_{\gamma_m,j}, X_{\gamma_m+1,j} | X_{\gamma_m,j})]$, $\underline{h}_{1,j}(X_{\gamma_m,j}) := \mathbb{E}[h(X_{\gamma_m-1,j}, X_{\gamma_m,j} | X_{\gamma_m,j})]$,
 $h_{2,j}(X_{\gamma_m+1,j}) := \mathbb{E}[\tilde{h}(X_{\gamma_m,j}, X_{\gamma_m+1,j} | X_{\gamma_m+1,j})]$, $\bar{h}_{1,j}(X_{\gamma_m+1,j}) := \mathbb{E}[h(X_{\gamma_m+1,j}, X_{\gamma_m+2,j} | X_{\gamma_m+1,j})]$,
 with $\theta_j^{(m)} := \mathbb{E}[h(X_{\gamma_m,j}, X_{\gamma_m+1,j})]$ being the signal jump of the j -th coordinate at γ_m . Note that by above definitions, $h_{1,j}(X_{\gamma_m,j})$ and $h_{2,j}(X_{\gamma_m+1,j})$ are the Hoeffding's projections for the two-sample kernel $h(x, y)$ with heterogeneous data and $\underline{h}_{1,j}(X_{\gamma_m,j})$, $\bar{h}_{1,j}(X_{\gamma_m+1,j})$ are the Hoeffding's projections for the two-sample kernel before and after the change point with homogeneous data. Moreover, define the four $\mathbb{R}^{s_m}$ -dimensional vectorized version of Hoeffding's projections as follows

$\mathbf{h}_1(\mathbf{X}_{\gamma_m}) = (h_{1,1}(X_{\gamma_m,1}), \dots, h_{1,s_m}(X_{\gamma_m,s_m}))^\top$, $\mathbf{h}_2(\mathbf{X}_{\gamma_m+1}) = (h_{2,1}(X_{\gamma_m+1,1}), \dots, h_{2,s_m}(X_{\gamma_m+1,s_m}))^\top$,
 $\bar{\mathbf{h}}_1(\mathbf{X}_{\gamma_m+1}) = (\bar{h}_{1,1}(X_{\gamma_m+1,1}), \dots, \bar{h}_{1,s_m}(X_{\gamma_m+1,s_m}))^\top$, $\underline{\mathbf{h}}_1(\mathbf{X}_{\gamma_m}) = (\underline{h}_{1,1}(X_{\gamma_m,1}), \dots, \underline{h}_{1,s_m}(X_{\gamma_m,s_m}))^\top$,
 and define the following four $\mathbb{R}^{s_m \times s_m}$ -dimensional covariance matrices as

$$\Sigma_1^{(m)} = \text{Cov}(\mathbf{h}_1(\mathbf{X}_{\gamma_m})), \quad \Sigma_2^{(m)} = \text{Cov}(\mathbf{h}_2(\mathbf{X}_{\gamma_m+1})), \\ \Sigma_3^{(m)} = \text{Cov}(\bar{\mathbf{h}}_1(\mathbf{X}_{\gamma_m+1}) - \mathbf{h}_2(\mathbf{X}_{\gamma_m+1})), \quad \Sigma_4^{(m)} = \text{Cov}(\underline{\mathbf{h}}_1(\mathbf{X}_{\gamma_m}) + \mathbf{h}_1(\mathbf{X}_{\gamma_m})).$$

To present the change point inference results, we need the following additional assumption.

Assumption (C.2): We require there exists constants $\kappa_1 > 0$ and $\kappa_2 > 0$ such that

$$\begin{aligned} \kappa_1 &\leq \min(\lambda_{\min}(\Sigma_1^{(m)}), \lambda_{\min}(\Sigma_2^{(m)}), \lambda_{\min}(\Sigma_3^{(m)}), \lambda_{\min}(\Sigma_4^{(m)}), \\ \max(\lambda_{\max}(\Sigma_1^{(m)}), \lambda_{\max}(\Sigma_2^{(m)}), \lambda_{\max}(\Sigma_3^{(m)}), \lambda_{\max}(\Sigma_4^{(m)}) &\leq \kappa_2, \text{ for } m = 1, \dots, M_0. \end{aligned} \quad (22)$$

With the above notations, we are ready to give the change point inference results.

THEOREM 5: *Suppose Assumptions (B.1)–(B.3) and (C.1) – (C.2) hold. Then we have*

$$\begin{aligned} (1) & |\tilde{\gamma}_m - \gamma_m| = O_P\left(\frac{1}{\|\boldsymbol{\theta}^{(m)}\|^2}\right), \text{ for } m = 1, \dots, M_0; \\ (2) & \|\boldsymbol{\theta}^{(m)}\|^2(\tilde{\gamma}_m - \gamma_m) \Rightarrow \arg \max_{s \in \mathbb{R}} Z^{(m)}(s), \text{ for } m = 1, \dots, M_0, \end{aligned} \quad (23)$$

with

$$Z^m(s) = \begin{cases} -s + \sigma_{1,*}^{(m)} W_1^{(m)}(s) + \sigma_{3,*}^{(m)} W_2^{(2)}(s) + \sigma_{2,*}^{(m)} W_3^{(m)}(s), & s > 0, \\ 0, & s = 0, \\ s + \sigma_{1,*}^{(m)} W_1^{(m)}(s) + \sigma_{4,*}^{(m)} W_2^{(m)}(s) + \sigma_{2,*}^{(m)} W_3^{(m)}(s), & s < 0, \end{cases}$$

where $\{W_1^{(m)}(s)\}$, $\{W_2^{(m)}(s)\}$, $\{W_3^{(m)}(s)\}$ are standard Brownian motion processes on $(-\infty, \infty)$

which are independent with each other and $\sigma_{j,*}^{(m)}$; $j = 1, 2, 3, 4$ are defined as

$$\begin{aligned} \sigma_{1,*}^{(m)} &= \frac{((\boldsymbol{\theta}^{(m)})^\top \Sigma_1^{(m)} \boldsymbol{\theta}^{(m)})^{1/2}}{\|\boldsymbol{\theta}^{(m)}\|}, \sigma_{2,*}^{(m)} = \frac{((\boldsymbol{\theta}^{(m)})^\top \Sigma_2^{(m)} \boldsymbol{\theta}^{(m)})^{1/2}}{\|\boldsymbol{\theta}^{(m)}\|}, \\ \sigma_{3,*}^{(m)} &= \frac{((\boldsymbol{\theta}^{(m)})^\top \Sigma_3^{(m)} \boldsymbol{\theta}^{(m)})^{1/2}}{\|\boldsymbol{\theta}^{(m)}\|}, \sigma_{4,*}^{(m)} = \frac{((\boldsymbol{\theta}^{(m)})^\top \Sigma_4^{(m)} \boldsymbol{\theta}^{(m)})^{1/2}}{\|\boldsymbol{\theta}^{(m)}\|}. \end{aligned} \quad (24)$$

Moreover, we can prove that $\|\boldsymbol{\theta}^{(m)}\|^2(\tilde{\gamma}_m - \gamma_m)$'s are asymptotically independent.

Theorem 5 (1) demonstrates that the U-statistic projection refinement algorithm U-PRA has an estimation error bound of $O_P(\frac{1}{\|\boldsymbol{\theta}^{(m)}\|^2})$, which has been improved compared to the initial estimators obtained in Theorem 3. To our best knowledge, the convergence rate is optimal. From the estimation point of view, the optimal rate of γ_m is a substantial enhancement compared to the initial estimator in Theorem 3. Moreover, this refinement is crucial from an inferential standpoint, as the key factor is the existence of an $O_P(\frac{1}{\|\boldsymbol{\theta}^{(m)}\|^2})$ rate, which ensures the presence of a limiting distribution. Lastly, the refined change point estimator itself achieves a convergence rate of $O_P(\frac{1}{\|\boldsymbol{\theta}^{(m)}\|^2})$, which remains independent of the dimensional parameters d , the sampling period n , and holds even in high-dimensional

settings or when the jump size is potentially diminishing. Theorem 5 (2) shows that the centered and scaled change point estimator, after refining, has an limiting distribution which follows the $\arg \max$ of a drifted weighted Brownian motion process. Hence, we can make valid confidence intervals for the detected change points. In Appdenx C, we provide more discussions and implementation details for making confidence interval.

5. Numerical performance

In this section, we examine the numerical performance of our proposed method in terms of change point detection, estimation as well as inference for the mean shift and variance change point models. We also compare our method with several existing state-of-art techniques. To save space, we put the detailed model settings and results in Web Appendix C of the supplementary materials. The simulation results can be summarized as follows:

For mean change point models, we consider two types of kernels: $h_1(x, y) = y - x$ and $h_2(x, y) = \text{sign}(y - x)$. The main findings are as follows:

1. **Size control:** When the error distribution is Gaussian, both methods effectively control the empirical size. However, under heavy-tailed errors or outlier contamination, the method based on $h_1(x, y)$ becomes overly conservative, while the h_2 -based method still maintains proper size control. See Table 1.

[Table 1 about here.]

2. **Multiple change point estimation:** When the data follow Gaussian distributions, the h_1 -based method outperforms both the h_2 -based and other mainstream approaches. However, under heavy-tailed errors, both the h_1 -based method and existing methods suffer from large estimation errors, while the h_2 -based method remains robust and accurately identifies both the number and locations of change points. Moreover, for both h_1 and h_2 , the refined change point estimators significantly improve over the initial estimators. See Table 2.

[Table 2 about here.]

3. Multiple change point inference: Under Gaussian errors, both h_1 and h_2 -based methods yield confidence intervals with empirical coverage probabilities close to the nominal level (95%). However, when data are heavy-tailed (e.g., t_3 or t_2 distributions) or contaminated with outliers, the coverage of the h_1 -based method drops substantially below the nominal level, while the h_2 -based method remains valid. See Table 3.

[Table 3 about here.]

4. Finally, for variance change point models, we also compare two kernels: $h_3(x, y) = y^2 - x^2$ and $h_4(x, y) = \text{sign}(y^2 - x^2)$, and observe similar patterns in the results. Comprehensive descriptions of the model specifications and corresponding numerical results are provided in Web Appendix D.

6. Real data analysis

In this section, we apply the newly proposed method to a genomic dataset consisting of bladder tumour profiles. The motivation for this real-world example is rooted in comparative genomic hybridization (CGH), a technique widely used in genomic studies to determine the gene copy number in DNA profiles. Alterations in the copy number, either an increase or decrease, are often associated with the presence of disease (Nicolas et al. (2006)). The dataset utilized in this analysis is sourced from the R package `ecp` and includes 43 DNA profiles ($d = 43$), each with a length of 2215 data points ($n = 2215$). As described by Matteson and James (2014), the dataset captures the relative hybridization intensity compared to a reference normal genome, with all data normalized on a logarithmic scale. The objective is to identify regions of DNA that exhibit abnormal copy numbers in the bladder tumour.

[Table 4 about here.]

In this data analysis, we employed two methods, $\mathbf{h}_1(\mathbf{x}, \mathbf{y}) = \mathbf{y} - \mathbf{x}$ and $\mathbf{h}_2(\mathbf{x}, \mathbf{y}) = \text{sign}(\mathbf{y} - \mathbf{x})$, for change point detection. According to Wang and Samworth (2018), the dataset contains a significant number of outliers; therefore, for $\mathbf{h}_1(\mathbf{x}, \mathbf{y})$, we first applied the

3- σ rule to truncate the data. We set the bandwidth to $G = 50$ and the number of bootstrap repetitions to $B=100$. Change point detection and estimation were carried out using the method described in Section 4. First, both methods successfully identified multiple change points in the data. Specifically, the h_1 -based method detected 30 change points, while the h_2 -based method identified 31 change points. Compared to Wang and Samworth (2018), which identifies 254 change points using the default threshold, our findings are more reasonable, as the use of bootstrap to determine critical values helps eliminate many false change points. Finally, we found that both the h_1 and h_2 methods detected 10 identical change points. We recorded the positions of these change points and their corresponding confidence intervals in Table 4. We can see that the confidence intervals for both methods are relatively short and show a high degree of overlap, which strongly indicates the accuracy of the estimated change points. Many of these change points have also been identified in existing studies. Figure 1 presents the change point detection results for the first three individuals, where we observe that the detected change points effectively partition the dataset.

[Figure 1 about here.]

7. Summary

This paper introduces a general framework for high-dimensional change-point detection, estimation, and inference based on U-statistics within a moving window. By enabling flexible kernel function selection, our method extends beyond traditional mean-based change-point analysis to accommodate a broader range of structural changes, including variance and robust mean shifts. We develop a minimax-optimal test statistic tailored for sparse alternatives and propose a high-dimensional multiplier bootstrap procedure that ensures valid significance control while maintaining high power. For estimation, we construct an initial estimator for the number and locations of change points and refine it using the U-statistic Projection Refinement Algorithm (U-PRA), achieving minimax-optimal localization rates. Furthermore,

we derive the asymptotic distribution of the refined estimators, facilitating the construction of valid confidence intervals. The effectiveness of our method is demonstrated through both theoretical analysis and empirical validation on synthetic and real-world datasets. Interesting future directions include extending the change-point analysis framework to other statistical settings, such as graphical models.

8. Supplementary Materials

Web Appendix contains the detailed proofs of theoretical results in Sections 3 and 4 and detailed numerical results in Section 5.

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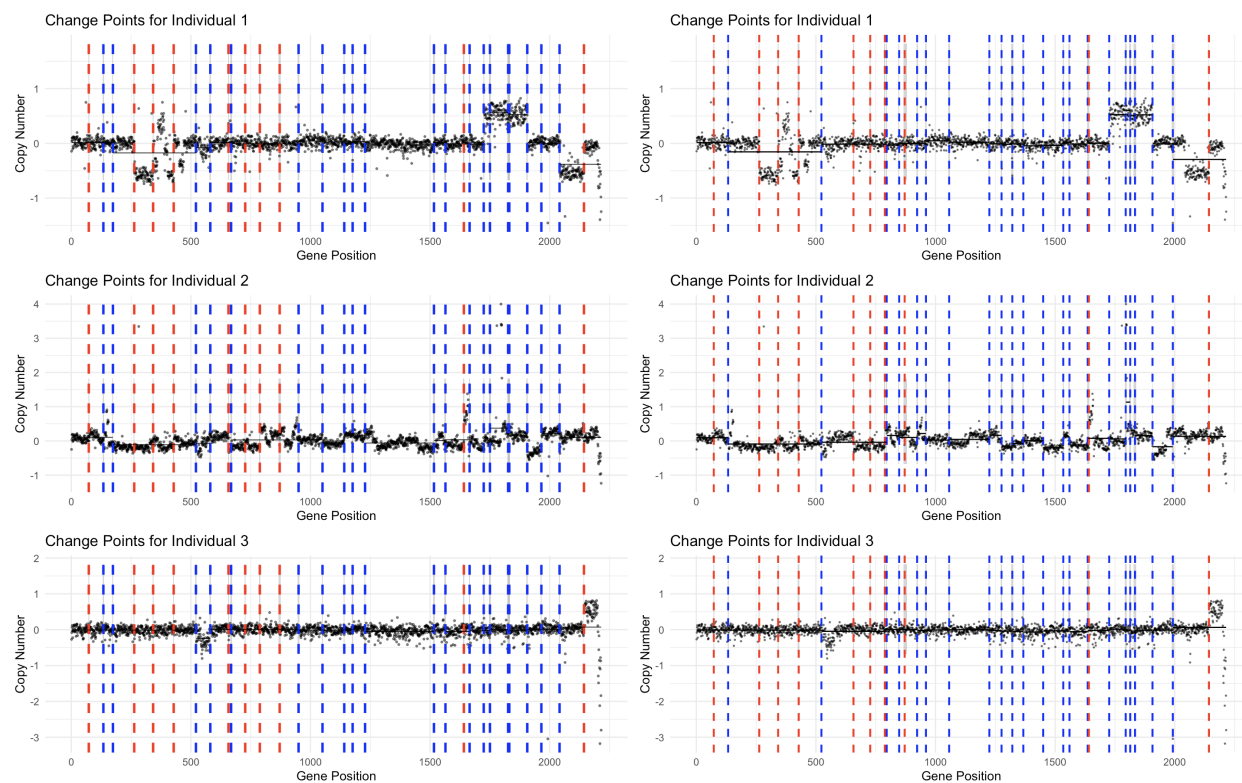


Figure 1: Change Point Detection Results for DNA Copy Number Variation for the method with $h_1(x, y) = y - x$ (left) and $h_2(x, y) = \text{sign}(y - x)$ (right). The red dashed lines correspond to the common locations detected by both methods.

Table 1: Empirical size performance under various model settings.

Model 1	$h_1(\mathbf{x}, \mathbf{y}) = \mathbf{y} - \mathbf{x}$			$h_2(\mathbf{x}, \mathbf{y}) = \text{sign}(\mathbf{y} - \mathbf{x})$		
	$d = 100$	$d = 200$	$d = 300$	$d = 100$	$d = 200$	$d = 300$
Normal	0.026	0.028	0.032	0.032	0.026	0.040
t_3	0.004	0.004	0.008	0.036	0.038	0.034
t_2	0	0	0	0.042	0.038	0.038
Contam-Norm	0.024	0.024	0.020	0.042	0.028	0.038
Model 2	$h_1(\mathbf{x}, \mathbf{y}) = \mathbf{y} - \mathbf{x}$			$h_2(\mathbf{x}, \mathbf{y}) = \text{sign}(\mathbf{y} - \mathbf{x})$		
	$d = 100$	$d = 200$	$d = 300$	$d = 100$	$d = 200$	$d = 300$
Normal	0.042	0.034	0.026	0.036	0.034	0.016
t_3	0.014	0.008	0.008	0.038	0.046	0.026
t_2	0	0.002	0	0.038	0.038	0.024
Contam-Norm	0.030	0.012	0.014	0.056	0.034	0.038

Table 2: Experimental results for multiple change points estimation under Models 1 and 2.

	Model 1				Model 2			
Normal	Haus	$\hat{M}_0 < M_0$	$\hat{M}_0 = M_0$	$\hat{M}_0 > M_0$	haus	$\hat{M}_0 < M_0$	$\hat{M}_0 = M_0$	$\hat{M}_0 > M_0$
$h_1 - initial$	5.965	0.000	0.985	0.015	5.760	0.000	0.985	0.015
$h_1 - refined$	1.360	0.000	0.985	0.015	1.465	0.000	0.985	0.015
$h_2 - initial$	5.645	0.000	0.975	0.025	6.135	0.000	0.965	0.035
$h_2 - refined$	1.405	0.000	0.975	0.025	1.330	0.000	0.965	0.035
LZZL	22.770	0.000	0.680	0.320	26.460	0.000	0.625	0.375
Inspect	15.565	0.005	0.725	0.270	20.200	0.005	0.625	0.370
t_3	Haus	$\hat{M}_0 < M_0$	$\hat{M}_0 = M_0$	$\hat{M}_0 > M_0$	haus	$\hat{M}_0 < M_0$	$\hat{M}_0 = M_0$	$\hat{M}_0 > M_0$
$h_1 - initial$	62.921	0.525	0.435	0.040	53.497	0.485	0.490	0.025
$h_1 - refined$	59.406	0.525	0.435	0.040	49.618	0.485	0.490	0.025
$h_2 - initial$	6.370	0.000	0.965	0.035	6.020	0.000	0.970	0.030
$h_2 - refined$	1.635	0.000	0.965	0.035	1.485	0.000	0.970	0.030
LZZL	30.125	0.065	0.700	0.235	66.060	0.290	0.550	0.160
Inspect	84.215	0.000	0.000	1.000	82.695	0.000	0.000	1.000
t_2	Haus	$\hat{M}_0 < M_0$	$\hat{M}_0 = M_0$	$\hat{M}_0 > M_0$	haus	$\hat{M}_0 < M_0$	$\hat{M}_0 = M_0$	$\hat{M}_0 > M_0$
$h_1 - initial$	171.143	1.000	0.000	0.000	135.889	0.995	0.005	0.000
$h_1 - refined$	171.714	1.000	0.000	0.000	135.444	0.995	0.005	0.000
$h_2 - initial$	8.995	0.005	0.920	0.075	11.325	0.025	0.925	0.050
$h_2 - refined$	2.940	0.005	0.920	0.075	5.950	0.025	0.925	0.050
LZZL	139.335	0.670	0.280	0.050	185.175	0.855	0.130	0.015
Inspect	87.310	0.000	0.000	1.000	87.035	0.000	0.000	1.000
Contam-Norm	Haus	$\hat{M}_0 < M_0$	$\hat{M}_0 = M_0$	$\hat{M}_0 > M_0$	haus	$\hat{M}_0 < M_0$	$\hat{M}_0 = M_0$	$\hat{M}_0 > M_0$
$h_1 - initial$	131.748	0.915	0.075	0.010	55.864	0.375	0.560	0.065
$h_1 - refined$	130.556	0.915	0.075	0.010	51.503	0.375	0.560	0.065
$h_2 - initial$	21.680	0.005	0.835	0.065	6.815	0.005	0.975	0.020
$h_2 - refined$	15.920	0.10	0.835	0.065	2.885	0.005	0.975	0.020
LZZL	116.605	0.575	0.285	0.140	132.185	0.665	0.190	0.145
Inspect	86.830	0.000	0.000	1.000	87.340	0.000	0.000	1.000

Table 3: Empirical confidence inference results under Model 1 and 2

Model 1	$\mathbf{h}_1(\mathbf{x}, \mathbf{y}) = \mathbf{y} - \mathbf{x}$			$\mathbf{h}_2(\mathbf{x}, \mathbf{y}) = \text{sign}(\mathbf{y} - \mathbf{x})$		
	γ_1	γ_2	γ_3	γ_1	γ_2	γ_3
Norm	0.935	0.925	0.900	0.945	0.945	0.910
t_3	0.805	0.756	0.756	0.945	0.955	0.925
t_2	-	-	-	0.940	0.950	0.935
Contam-Norm	0.677	0.694	0.743	0.942	0.933	0.922
Model 2	γ_1	γ_2	γ_3	γ_1	γ_2	γ_3
	γ_1	γ_2	γ_3	γ_1	γ_2	γ_3
Norm	0.925	0.895	0.920	0.935	0.915	0.950
t_3	0.786	0.830	0.803	0.960	0.920	0.955
t_2	-	-	-	0.929	0.924	0.930
Contam-Norm	0.753	0.753	0.750	0.925	0.895	0.960

Table 4: Common Change Point Detection Results for DNA Copy Number Variation

Change Point	$\mathbf{h}_1(\mathbf{x}, \mathbf{y}) = \mathbf{y} - \mathbf{x}$		$\mathbf{h}_2(\mathbf{x}, \mathbf{y}) = \text{sign}(\mathbf{y} - \mathbf{x})$	
	Lower Bound	Upper Bound	Lower Bound	Upper Bound
73	71	75	71	76
263	261	264	261	264
342	340	343	340	343
428	425	429	424	430
657	655	659	655	660
727	724	728	725	729
788	784	790	785	790
871	866	875	867	880
1641	1638	1642	1638	1642
2143	2141	2144	2141	2144